Covalent Bonds

Exam code: 0620 & 0971

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The formation of covalent bonds

Covalent compounds

- Covalent compounds are formed when **pairs** of electrons are **shared** between atoms
- Only **non-metal elements** participate in covalent bonding
- As in ionic bonding, each atom gains a **full outer shell** of electrons, giving them a noble gas electronic configuration
- Covalently bonded substances may consist of small molecules or giant molecules
- When two or more atoms are **covalently** bonded together, we describe them as '**molecules**'
- **Dot-and-cross** diagrams can be used to show the electronic configurations in simple molecules
  - Electrons from one atom are represented by a dot, and the electrons of the other atom are represented by a cross
  - The electron shells of each atom in the molecule overlap and the shared electrons are shown in the area of overlap
  - The dot-and-cross diagram of the molecule shows clearly which atom each electron originated from

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Diagram to show the formation of a covalent bond

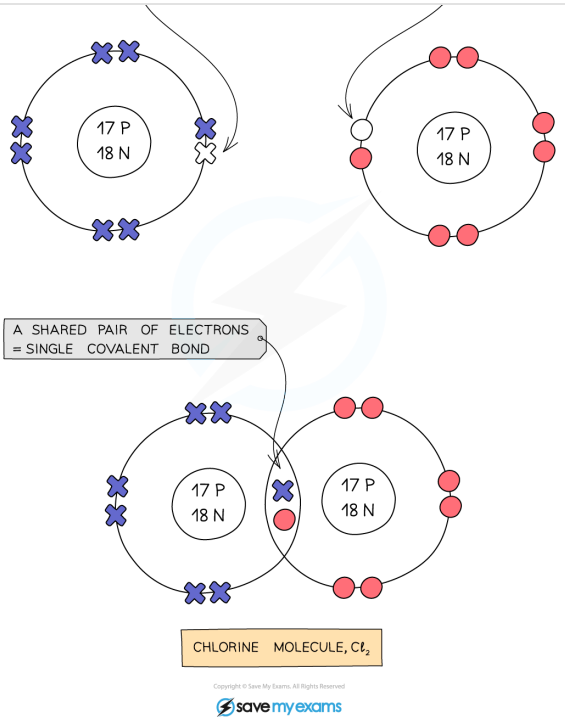


Diagram showing how a covalent bond forms between two chlorine atoms

Examiner Tips and Tricks

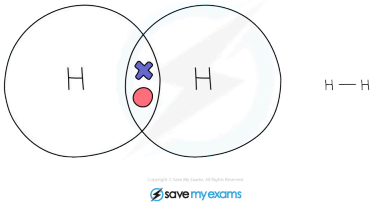
When drawing dot-and-cross diagrams for covalent compounds, make sure that the electron shell for each atom is full (remember that the 1st shell can only hold 2 electrons).

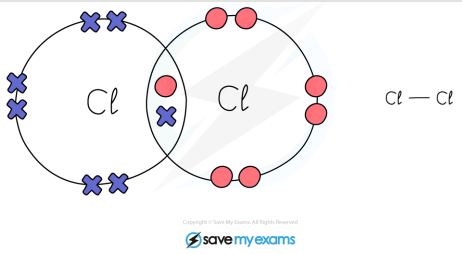
Single covalent bonds

- Many simple molecules exist in which two adjacent atoms share one pair of electrons, also known as a single covalent bond (or single bond)

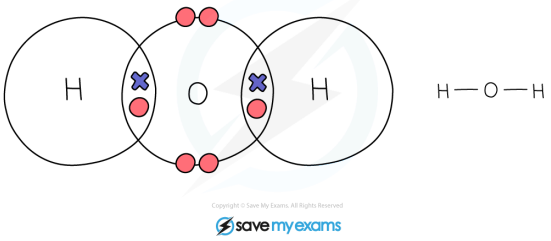
Common Examples of Simple Molecules

Hydrogen:

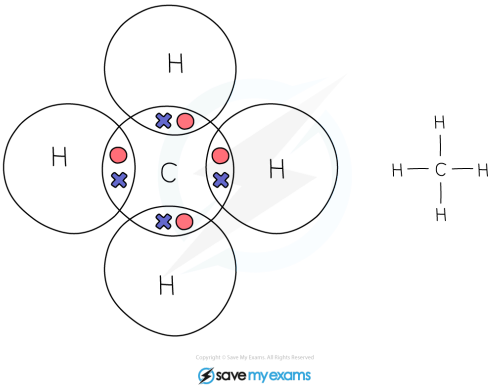




Water:



Methane:



Ammonia:

Hydrogen chloride:

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